

VPA ICT — Year 8: Cybersecurity, Robotics & Media

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Duration: 36 weeks | Hardware: micro:bit v2, Arduino starter kits | Software: Python, MakeCode, OpenShot/Clipchamp

Unit 1: Cybersecurity Awareness (Weeks 1 – 9)

Week 1 — Introduction to Cybersecurity

Keywords: cybersecurity 网络安全 (wǎngluò ānquán), threat 威胁 (wēixié), vulnerability 漏洞 (lòudòng), attack 攻击 (gōngjī), defense 防御 (fángyù)

Activity 1 (15 min): "Hacker vs. Defender" role-play. Students draw cards: phishing email, strong password, malware, firewall.

Activity 2 (15 min): Case studies — Real news stories about data breaches (simplified). Discuss: What went wrong?

Activity 3 (10 min): Class brainstorming — "What devices in your home connect to the internet?" (IoT awareness)

Week 2 — Password Security

Keywords: password 密码 (mìmǎ), encryption 加密 (jiāmì), brute force 暴力破解 (bàolì pòjiě), two-factor authentication 双因素认证 (shuāng yīnsù rènzhèng), passphrase 密码短语 (mìmǎ duǎnyǔ)

Activity 1 (15 min): Password strength tester — Students test passwords on howsecureismypassword.net (safe demo).

Activity 2 (15 min): Create a strong passphrase — "CorrectHorseBatteryStaple" concept. Memorable but long.

Activity 3 (10 min): Password manager introduction — Why reuse is dangerous.

Week 3 — Phishing & Social Engineering

Keywords: phishing 网络钓鱼 (wǎngluò diàoyú), spoofing 欺骗 (qīpiàn), social engineering 社会工程学 (shèhuì gōngchéngxué), suspicious 可疑的 (kěyí de), verify 核实 (héshí)

Activity 1 (15 min): "Spot the Phish" — Show 5 emails. Which are real? Which are phishing? Red flags: urgency, weird links, spelling errors.

Activity 2 (15 min): URL analysis — Compare amazon.com vs. amazon-security-verify.com

Activity 3 (10 min): Create a "Phishing Warning" poster with 5 red flags.

Week 4 — Malware & Viruses

Keywords: malware 恶意软件 (èyì ruǎnjiàn), virus 病毒 (bìngdú), trojan 木马 (mùmǎ), ransomware 勒索软件 (lèsuǒ ruǎnjiàn), antivirus 杀毒软件 (shādú ruǎnjiàn)

Activity 1 (15 min): Malware types matching game — Match type to description and example.

Activity 2 (15 min): "How do I protect myself?" — Install updates, use antivirus, don't download from unknown sources.

Activity 3 (10 min): Discussion — "Can phones get viruses too?"

Week 5 — Network Security Basics

Keywords: firewall 防火墙 (fánghuǒqiáng), router 路由器 (lùróuqì), WPA3 WPA3加密协议 (WPA3 jiāmì xiéyì), SSID 网络名称 (wǎngluò míngchēng), encryption 加密 (jiāmì)

Activity 1 (15 min): Diagram a home network — Router connects to phone, laptop, smart TV. Where are the vulnerabilities?

Activity 2 (15 min): "Secure Your Home WiFi" — Change default password, hide SSID, enable WPA3.

Activity 3 (10 min): Public WiFi dangers — Why is free airport WiFi risky?

Week 6 — Cryptography Introduction

Keywords: cryptography 密码学 (mímǎ xué), cipher 密码 (mímǎ), Caesar cipher 凯撒密码 (kǎisā mímǎ), encryption key 加密密钥 (jiāmì mìyùè), plaintext 明文 (míngwén), ciphertext 密文 (mìwén)

Activity 1 (20 min): Caesar cipher workshop — Shift each letter by 3. Encode and decode messages.

Activity 2 (15 min): Python cipher — Write a simple encryption program.

```
message = "HELLO"
```

```
key = 3
```

```
encrypted = ""
```

```
for char in message:
```

```
    encrypted += chr(ord(char) + key)
```

```
print(encrypted)
```

Activity 3 (5 min): Discussion — Why is Caesar cipher not secure today?

Week 7 — Ethical Hacking & Penetration Testing

Keywords: ethical hacker 白帽黑客 (báimào hēikè), penetration test 渗透测试 (shèntòu cèshì), white hat 白帽 (báimào), black hat 黑帽 (hēimào), gray hat 灰帽 (huīmào)

Activity 1 (15 min): Ethical discussion — "Is it ever OK to hack?" White hat vs. black hat scenarios.

Activity 2 (15 min): "Bug Bounty" concept — Companies pay hackers to find vulnerabilities.

Activity 3 (10 min): Research one famous ethical hacker (e.g., Kevin Mitnick's rehabilitation).

Week 8 — Digital Privacy & Laws

Keywords: privacy 隐私 (yǐn sī), GDPR 通用数据保护条例 (tōngyòng shùjù bǎohù tiáolì), consent 同意 (tóngyì), data protection 数据保护 (shùjù bǎohù), cookie Cookie (网站追踪数据)

Activity 1 (15 min): "What do websites know about you?" Check browser cookies and permissions.

Activity 2 (15 min): Create a "Privacy Checklist" for students: review app permissions, turn off location, read privacy policies (simplified).

Activity 3 (10 min): Debate — "Should schools be allowed to monitor student internet use?"

Week 9 — Cybersecurity Assessment

Project (40 min): Create a "Student Cybersecurity Guide" (1-page infographic or 3-slide presentation) covering:

3 types of cyber threats

5 ways to stay safe online

What to do if hacked

One real case study

Unit 2: Robotics & Physical Computing (Weeks 10 – 18)

Week 10 — Introduction to micro:bit

Keywords: micro:bit 微型电脑板 (micro:bit 微型电脑板), LED matrix LED点阵 (LED diǎnzhèn), sensor 传感器 (chuángǎnqì), GPIO 通用输入输出 (tōngyòng shūrù shūchū), emulator 模拟器 (mónǐqì)

Activity 1 (15 min): micro:bit hardware tour — Identify buttons, LED grid, pins, USB, battery connector.

Activity 2 (15 min): First program on MakeCode — Display name scrolling across LED matrix.

Activity 3 (10 min): Download to physical micro:bit (if available) or test in emulator.

Week 11 — Buttons & Inputs

Activity 1 (15 min): "Rock Paper Scissors" — Press A for rock, B for paper, A+B for scissors. micro:bit randomly chooses.

Activity 2 (15 min): Counter program — A button adds 1, B button subtracts 1. Display count.

Activity 3 (10 min): "Reaction Game" — micro:bit shows icon; first to press button wins.

Week 12 — Sensors: Accelerometer & Compass

Keywords: accelerometer 加速度计 (jiāosùdùjì), compass 指南针 (zhǐnánzhēn), tilt 倾斜 (qīngxié), gesture 手势 (shǒushì), shake 摇晃 (yáohuàng)

Activity 1 (15 min): "Magic 8-Ball" — Shake micro:bit to get random answer.

Activity 2 (15 min): "Step Counter" — Use shake detection to count steps.

Activity 3 (10 min): Compass app — micro:bit points to North when button pressed.

Week 13 — Sensors: Temperature & Light

Activity 1 (15 min): Room temperature monitor — Display current temperature in °C.

Activity 2 (15 min): "Night Light" — LED brightness adjusts based on ambient light level.

Activity 3 (10 min): Data logging — Record temperature every minute for 10 minutes; analyze trend.

Week 14 — Radio Communication

Keywords: radio 无线电 (wúxiàn diàn), broadcast 广播 (guǎngbō), receive 接收 (jiēshōu), group 组 (zǔ), signal 信号 (xìnhào)

Activity 1 (15 min): "Send a Message" — Program two micro:bits to send text messages to each other.

Activity 2 (15 min): "Fire Alarm Network" — One micro:bit detects heat (simulated with button); sends alert to all others.

Activity 3 (10 min): Discussion — How does this relate to real IoT networks?

Week 15 — Introduction to Arduino

Keywords: Arduino 开源电子平台 (kāiyuán diànzǐ píngtái), breadboard 面包板 (miànbāo bǎn), resistor 电阻 (diànzǔ), LED 发光二极管 (fāguāng èrjíguǎn), jumper wire 跳线 (tiàoxiàn)

Activity 1 (15 min): Arduino hardware identification — Uno board, digital pins, analog pins, power, ground.

Activity 2 (20 min): "Blink" sketch — Upload the classic LED blink program. Change blink speed.

Activity 3 (5 min): Circuit safety — Never connect 5V directly to ground!

```
void setup() {  
  pinMode(LED_BUILTIN, OUTPUT);  
  
  void loop() {  
    digitalWrite(LED_BUILTIN, HIGH);  
  
    delay(1000);  
  
    digitalWrite(LED_BUILTIN, LOW);  
  
    delay(1000);  
  }  
}
```

Week 16 — Arduino: Inputs & Outputs

Activity 1 (20 min): "Traffic Light" — Connect 3 LEDs (red, yellow, green) with resistors. Program traffic light sequence.

Activity 2 (15 min): Add a button — Pedestrian crossing: press button, lights change.

Activity 3 (5 min): Debugging — "My LED won't light up!" Checklist: resistor? correct pin? power?

Week 17 — Arduino: Analog & PWM

Keywords: analog 模拟 (mónǐ), digital 数字 (shùzì), PWM 脉宽调制 (màikuān tiáozhì), potentiometer 电位器 (diànwèiqì), fade 渐变 (jiànbàn)

Activity 1 (15 min): LED fading — Use PWM to make LED slowly brighten and dim.

Activity 2 (15 min): Potentiometer control — Turn knob to control LED brightness.

Activity 3 (10 min): Analog read — Display potentiometer value on Serial Monitor.

Week 18 — Robotics Assessment

Project (40 min): Build and program one of the following:

Alarm System: micro:bit with motion sensor triggers LED and radio alert

Mood Lamp: Arduino with RGB LED changes color based on button presses

Digital Dice: micro:bit displays random number 1–6 when shaken

Submit: working prototype + 1-paragraph explanation of how it works.

Unit 3: Digital Media & Data Visualization (Weeks 19 – 27)

Week 19 — Video Editing Basics

Keywords: timeline 时间线 (shíjiān xiàn), clip 片段 (piànduàn), trim 修剪 (xiūjiǎn), split 分割 (fēngē), transition 转场 (zhuǎnchǎng), export 导出 (dǎochū)

Activity 1 (15 min): OpenShot/Clipchamp interface tour — Timeline, preview, media library.

Activity 2 (15 min): Import 3 video clips. Arrange on timeline. Trim beginning and end.

Activity 3 (10 min): Add a simple crossfade transition between clips.

Week 20 — Titles, Captions & Effects

Activity 1 (15 min): Add animated title at start of video. Add credits at end.

Activity 2 (15 min): Add subtitles/captions to a 30-second clip.

Activity 3 (10 min): Green screen chroma key — Replace green background with image.

Week 21 — Audio Editing

Activity 1 (15 min): Add background music. Adjust volume so voice is clear.

Activity 2 (15 min): Remove background noise using audio filters.

Activity 3 (10 min): Record voiceover narration for a video.

Week 22 — Data Visualization Principles

Keywords: data visualization 数据可视化 (shùjù kěshìhuà), infographic 信息图 (xìnxī tú), dashboard 仪表盘 (yíbiǎopán), misleading graph 误导性图表 (wùdǎo xìng túbiǎo), trend 趋势 (qūshì)

Activity 1 (15 min): "Good vs. Bad Charts" analysis — Show examples. Discuss: truncated y-axis, pie charts with too many slices, 3D distortions.

Activity 2 (15 min): Redesign a misleading chart to be honest and clear.

Activity 3 (10 min): Choose the right chart type: bar for comparison, line for trend, pie for proportion.

Week 23 — Python Data Visualization

Activity 1 (20 min): Introduction to matplotlib. Create a simple bar chart:

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr"]
sales = [120, 150, 180, 170]
plt.bar(months, sales)
plt.title("Monthly Sales")
plt.show()
```

Activity 2 (15 min): Create line chart, pie chart, and scatter plot from class survey data.

Activity 3 (5 min): Save charts as PNG files for use in presentations.

Week 24 — Infographic Design

Activity 1 (30 min): Design an infographic about a school topic (recycling, healthy eating, digital safety) using Canva or PowerPoint.

Requirements: Title, 3 data visualizations, icons, color scheme, sources cited.

Activity 2 (10 min): Peer review using design principles checklist.

Week 25 — Cloud Computing & Collaboration

Keywords: cloud 云 (yún), collaboration 协作 (xiézuò), sync 同步 (tóngbù), version control 版本控制 (bǎnběn kòngzhì), Google Workspace Google工作空间 (Google gōngzuò kōngjiān)

Activity 1 (15 min): Google Workspace collaboration — Simultaneous editing of a document. Watch cursors move!

Activity 2 (15 min): Version history — Show how to restore previous versions. "Undo" for entire document.

Activity 3 (10 min): Cloud storage comparison — Google Drive vs. OneDrive vs. iCloud.

Week 26 — Review & IGCSE Prep Introduction

Activity 1 (20 min): Mixed skills challenge — Document, spreadsheet, and presentation tasks in one lesson.

Activity 2 (15 min): IGCSE ICT (0417) overview — What to expect in Grades 9–10.

Activity 3 (5 min): Self-assessment: "Am I ready for IGCSE ICT?"

Week 27 — End of Year Assessment

Portfolio Review + Practical (40 min):

Cybersecurity guide (submitted earlier)

Robotics project demo (3 min per student)

Video editing project (1-minute edited video)

Data visualization (Python chart or infographic)

Reflection: "How has ICT helped me become a better learner?" (written, 100 words)

Standards Mapping

Unit	Standard	Description
Cybersecurity	CSTA 2-IC-22	Explain how physical and digital security measures protect electronic information
Robotics	CSTA 2-AP-12	Design and iteratively develop programs for practical intent
Media	ISTE 1.6	Communicate clearly and express themselves creatively using digital media
Data Viz	CCSS. MATH. 8. SP. A. 1	Construct and interpret scatter plots for bivariate measurement data